C - Intro

Why Learn C?

- help develop better understanding of what higher (and lower) level languages are doing
- help understand memory management
- system programming - interact with the OS

Compiled vs Interpreted Languages

- Compiled:
  - C (gcc program_file.c)
  - Java (javac program_file.java)
- Interpreted: can simply run without explicitly compiling
  - Bash (bash name_of_program.sh)
  - Python (python name_of_program.py)
  - Awk (gawk name_of_program.awk)

C vs. Other Languages

- Portability:
  - C = less portable, compiled code is architecture dependent
  - Java = more portable, compiled byte code runs the same on same JVM version
  - Python = more portable, mostly depends on python version, though underlying packages may be compiled underneath

C vs. Other Languages (continued)

- Speed
  - C is fast
  - Java is slower, but can be reasonably fast (JIT can make optimizations)
  - Python is known for being slow, but can be fast
    - speed comes when using libraries implemented underneath in C

Getting Started with C

- Hello World

```c
#include <stdio.h>
int main(void) {
    printf("Hello World\n")
    return 0;
}
```

C Preprocessor

- Performs source code substitution
- # indicates preprocessing directive
- include indicates preprocessing directive
  - typically use to include header files (we'll talk about what these mean later)
  - kind of like "importing"
- define PI 3.14159
  - everywhere PI occurs, replace with value
  - typically uses all caps

Running C Code

- Compiling
  - gcc program_file.c
  - gcc -o output_name program_file.c
- Running
  - ./a.out
  - ./output_name
- NOT THIS
  - gcc -o project.c project.c